

Project Design Phase-II Technology Stack (Architecture & Stack)

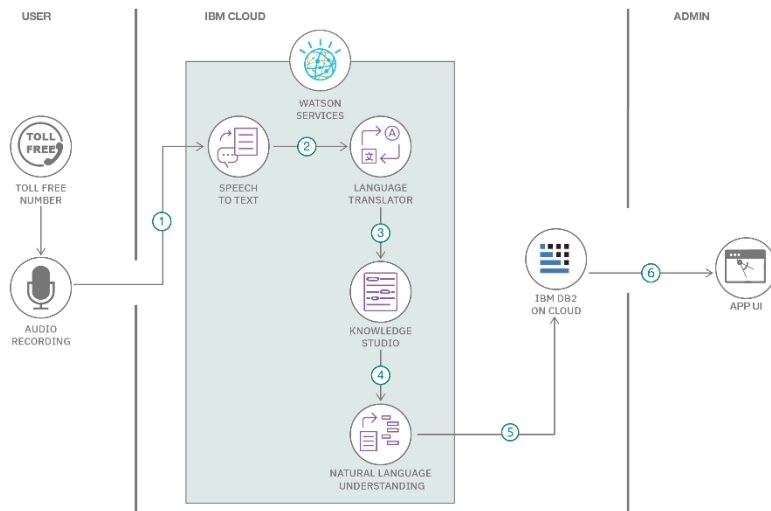
Date	18 June 2026
Team ID	
Project Name	ShopEZ
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: Order processing during pandemics for offline mode

Reference: <https://developer.ibm.com/patterns/ai-powered-backend-system-for-order-processing-during-pandemics/>



Guidelines:

- Include all the processes (As an application logic / Technology Block)
- Provide infrastructural demarcation (Local / Cloud)
- Indicate external interfaces (third party API's etc.)
- Indicate Data Storage components / services
- Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Fast, adaptive Single Page Application (SPA) for both the customer storefront and admin dashboard.	React.js, Tailwind CSS, Vite.
2.	Application Logic-1	Core backend logic providing RESTful routes to handle products, users, and checkout processes.	Node.js, Express.js.
3.	Application Logic-2	Client-side state management for maintaining persistent carts and wishlists across sessions.	React Context API, Browser LocalStorage.
4.	Application Logic-3	React Context API, Browser LocalStorage.	JSON Web Tokens (JWT), bcrypt.
5.	Database	Object Data Modeling (ODM) layer to enforce schema validation for users, products, and orders.	Mongoose ODM.
6.	Cloud Database	Fully managed, scalable NoSQL database utilizing replica sets for high availability.	MongoDB Atlas.
7.	File Storage	Storage mechanism for frontend static assets and dynamically uploaded product images.	Vercel Edge Network, Cloudinary API (Standard for MERN image hosting).
8.	External API-1	Secure payment processing gateway to handle customer checkout transactions.	Stripe API / Razorpay API.
9.	Machine Learning Model	Algorithm logic for the "similar products" recommendation strip.	Category-matching logic (JavaScript).
10.	Infrastructure (Server / Cloud)	Decoupled cloud deployments. Frontend static assets distributed globally; backend REST API hosted on stateless containers.	Vercel (Frontend), Render / AWS (Backend).

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	The core open-source libraries driving the application's user interface and server environment.	React.js, Node.js, Express.js, Tailwind CSS.
2.	Security Implementations	Cryptographic password hashing, token-based endpoint protection, and role-based access control preventing unauthorized admin access.	bcrypt (SHA-based hashing), JWT authorization headers.
3.	Scalable Architecture	Decoupled architecture separating the UI from the backend, utilizing stateless API executions for easy horizontal scaling.	MERN Stack, RESTful APIs, MongoDB Sharding.
4.	Availability	System achieves high uptime by leveraging managed cloud infrastructure that supports automated failovers.	Vercel Edge CDN, MongoDB Atlas (Replica Sets).
5.	Performance	Optimized frontend bundle sizes (under 300KB) through tree-shaking and rapid backend API response times (under 500ms).	Vite Bundler, Tailwind CSS utilities.

References:

<https://c4model.com/>

<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>

<https://www.ibm.com/cloud/architecture>

<https://aws.amazon.com/architecture>

<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>

